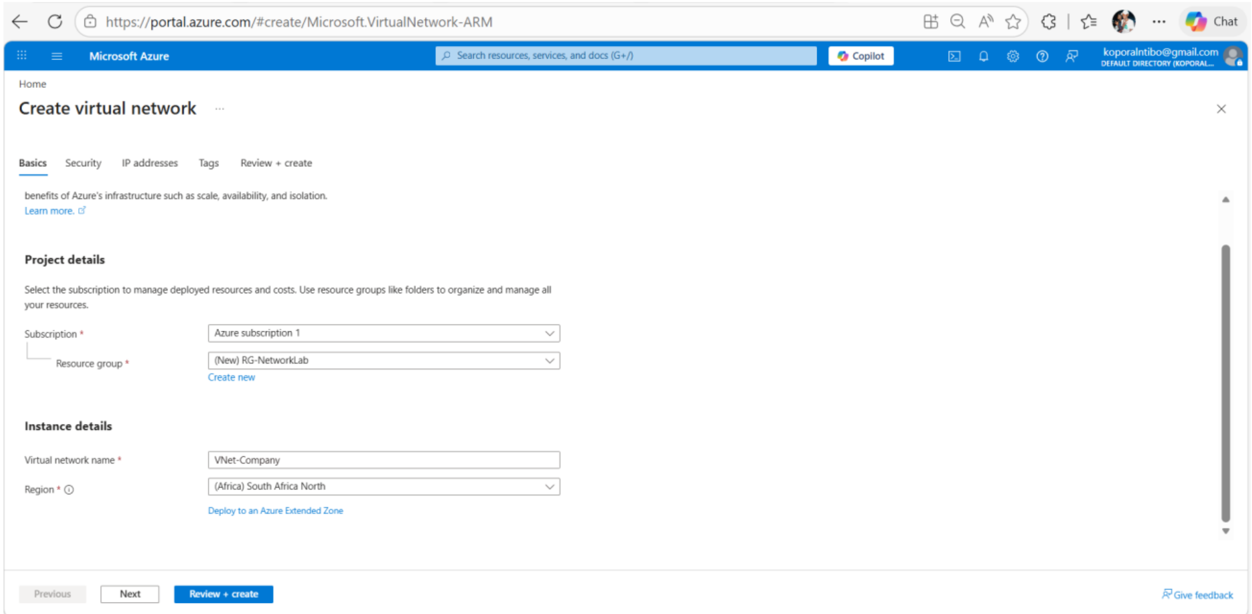
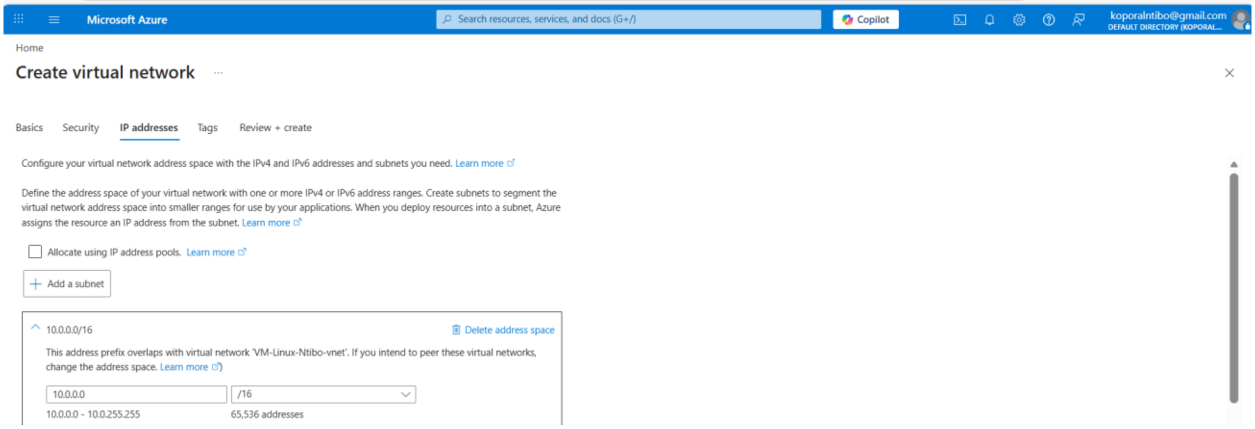
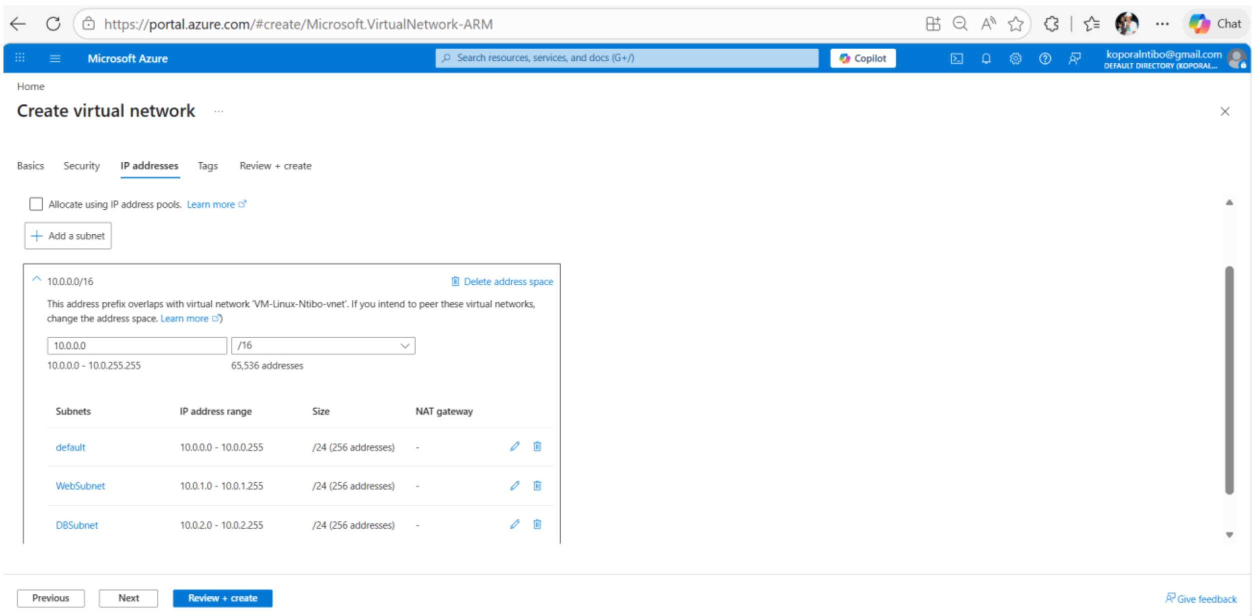
Chapter 3 Lab: Designing and Securing an Azure Virtual Network

Task 1: Create a Virtual Network (VNet)



I created a new resource group called **"RG-NetworkLab"** using **one Azure subscription** to manage and organize all resources within the environment. This resource group acts as a container for related resources, simplifying management, monitoring, and access control, while allowing easy deployment or deletion of all associated resources. Within this group, I deployed a **Virtual Network (VNet)**, which serves as the foundational building block for networking in Azure. The VNet enables virtual machines and services to communicate securely with each other, connect to the internet, and integrate with on-premises networks, while providing isolation, scalability, and segmentation similar to a traditional data center network. The region selected was **South Africa North**, as it is geographically close, reducing latency, ensuring compliance, and optimizing performance and cost. This setup provides a secure, organized, and scalable network foundation for all lab resources.


Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
koporalntibo@gmail.com
DEFAULT DIRECTORY (KOPORAL...
Home
Create virtual network
Basics
Security
IP addresses
Tags
Review + create
Configure your virtual network address space with the IPv4 and IPv6 addresses and subnets you need. Learn more
Define the address space of your virtual network with one or more IPv4 or IPv6 address ranges. Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet. Learn more
Allocate using IP address pools. Learn more
Add a subnet
10.0.0.0/16
Delete address space
This address prefix overlaps with virtual network 'VM-Linux-Ntibo-vnet'. If you intend to peer these virtual networks, change the address space. Learn more
10.0.0.0
/16
10.0.0.0 - 10.0.255.255
65,536 addresses


https://portal.azure.com/#create/Microsoft.VirtualNetwork-ARM
Chat
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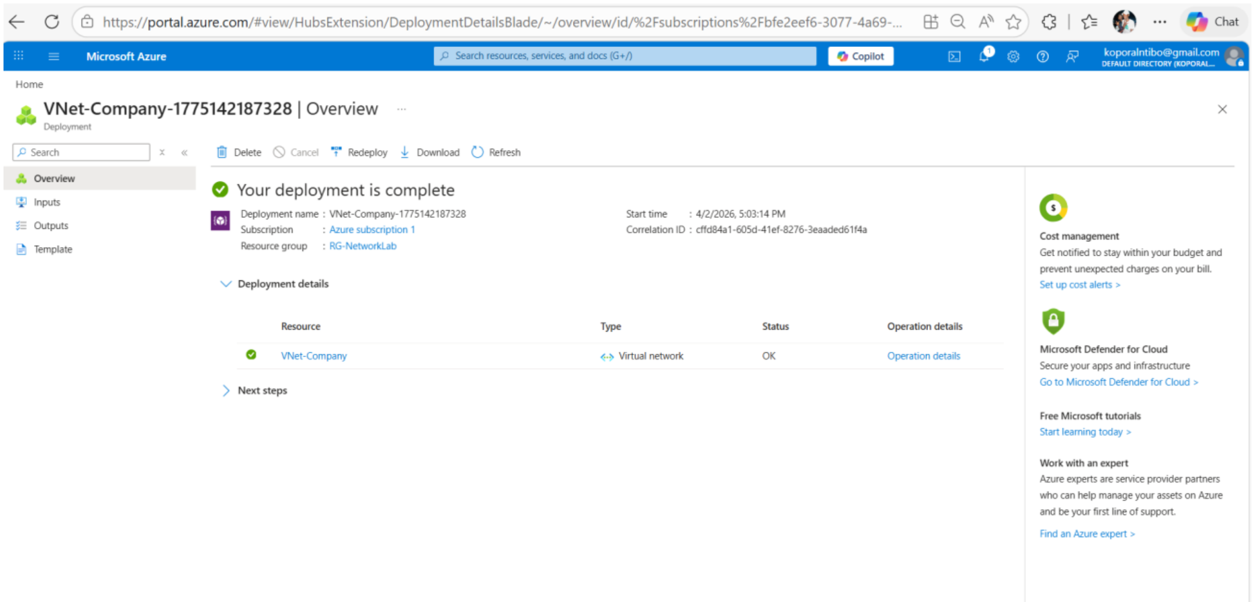
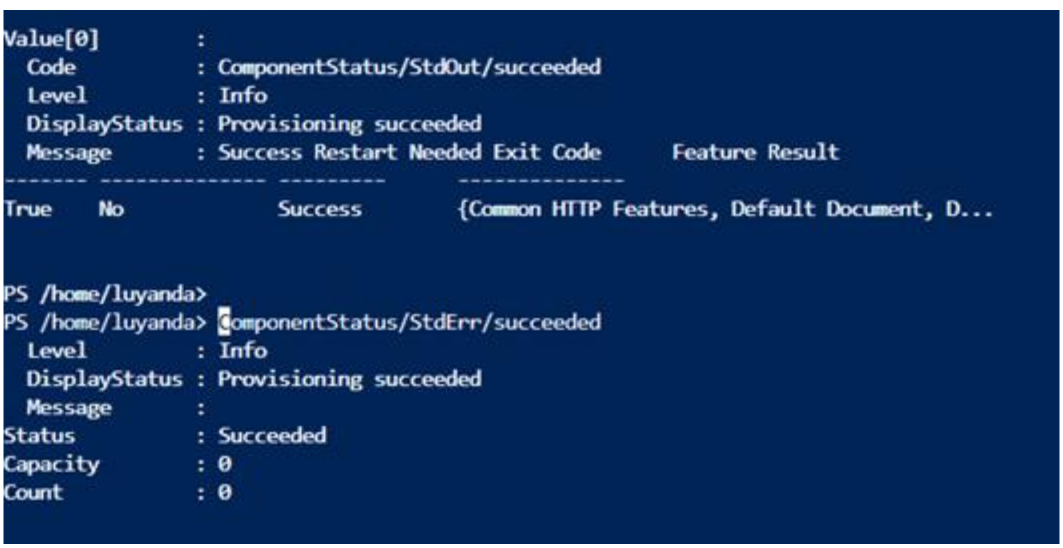
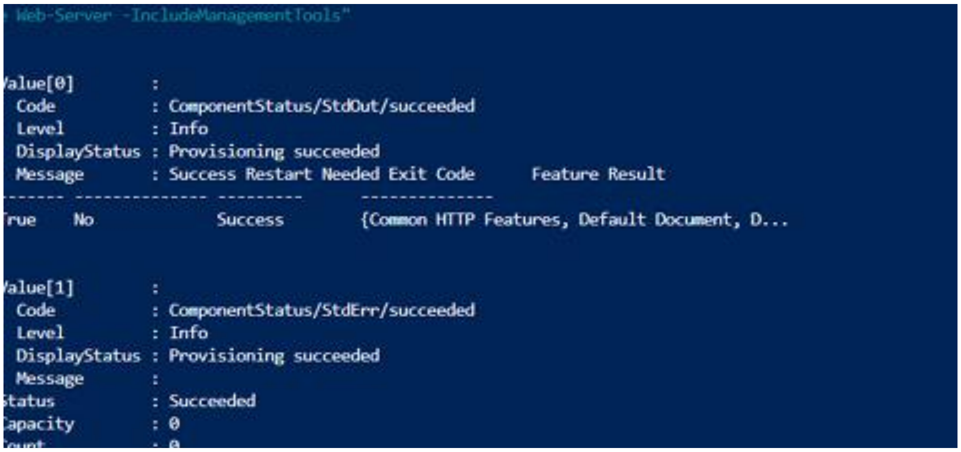
10.0.0.0
/16
10.0.0.0 - 10.0.255.255
65,536 addresses
Subnets
IP address range
Size
NAT gateway
default
10.0.0.0 - 10.0.0.255
/24 (256 addresses)
-
WebSubnet
10.0.1.0 - 10.0.1.255
/24 (256 addresses)
-
DBSubnet
10.0.2.0 - 10.0.2.255
/24 (256 addresses)
-
Previous
Next
Review + create
Give feedback

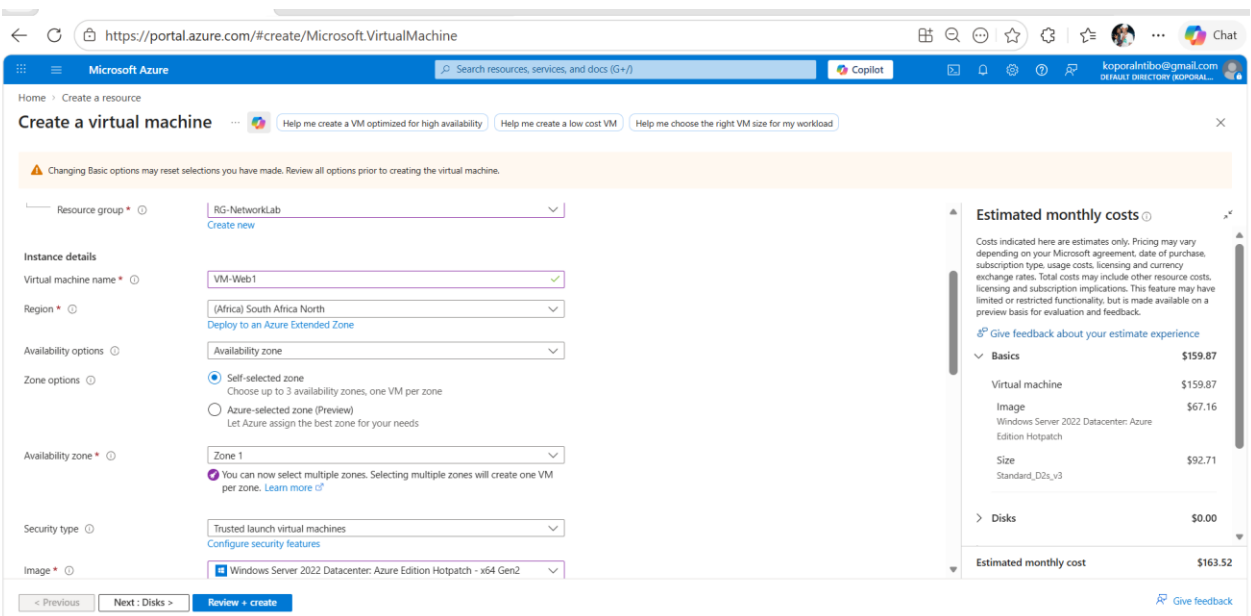

https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2Fbfe2eef6-3077-4a69-...
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
koporalntibo@gmail.com
Home
VNet-Company-1775142187328 | Overview
Deployment
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Your deployment is complete
Deployment name : VNet-Company-1775142187328
Subscription : Azure subscription 1
Resource group : RG-NetworkLab
Start time : 4/2/2026, 5:03:14 PM
Correlation ID : cffd84a1-605d-41ef-8276-3eaaded61f4a
Deployment details
Resource
Type
Status
Operation details
VNet-Company
Virtual network
OK
Operation details
Next steps
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >

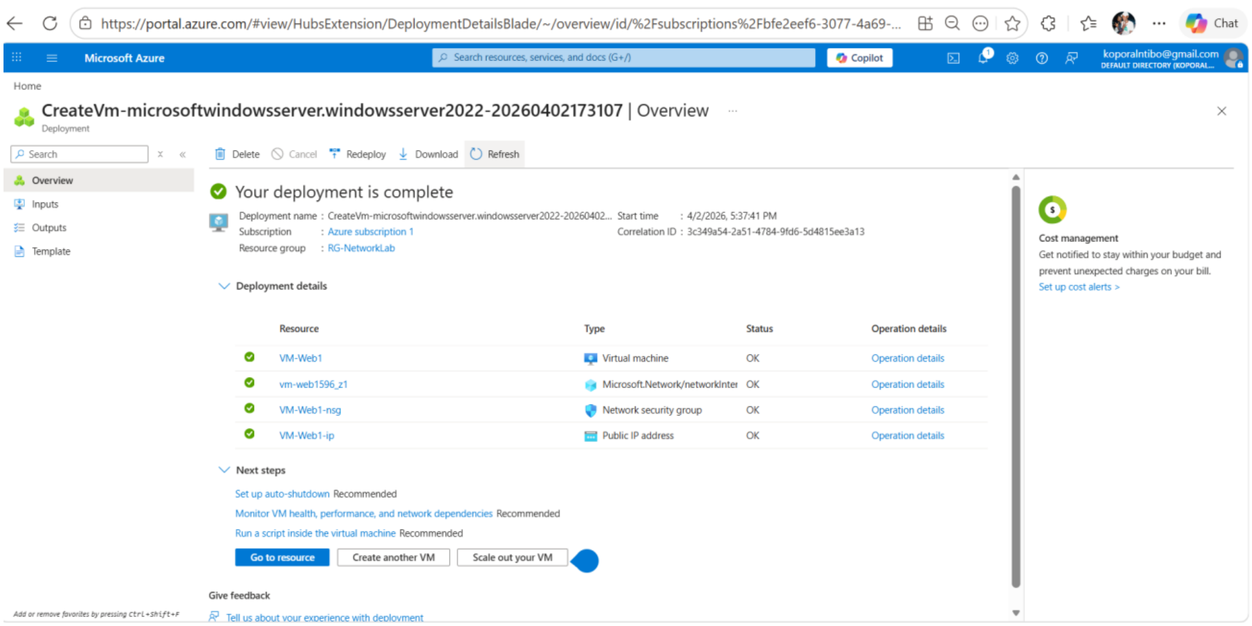

Value[0]   :
  Code   : ComponentStatus/StdOut/succeeded
  Level   : Info
  DisplayStatus : Provisioning succeeded
  Message   : Success Restart Needed Exit Code   Feature Result
True   No   Success   {Common HTTP Features, Default Document, D...
PS /home/luyanda>
PS /home/luyanda> ComponentStatus/StdErr/succeeded
  Level   : Info
  DisplayStatus : Provisioning succeeded
  Message   :
Status   : Succeeded
Capacity   : 0
Count   : 0

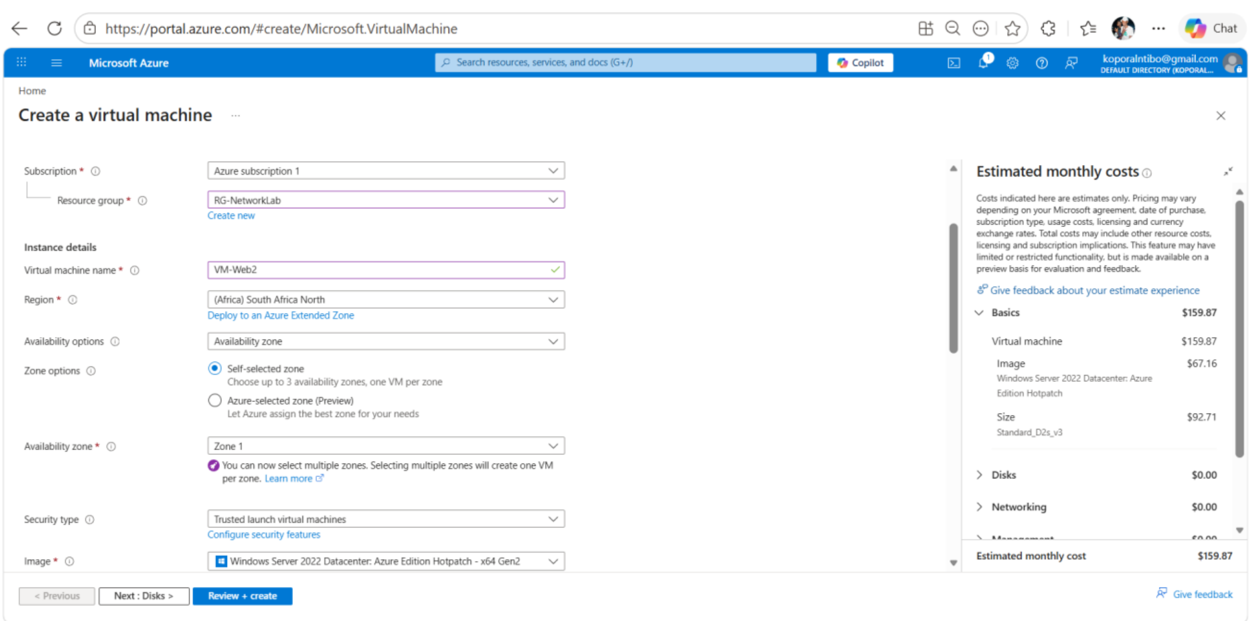


Task 2: Deploy Two Web Servers in Web Subnet

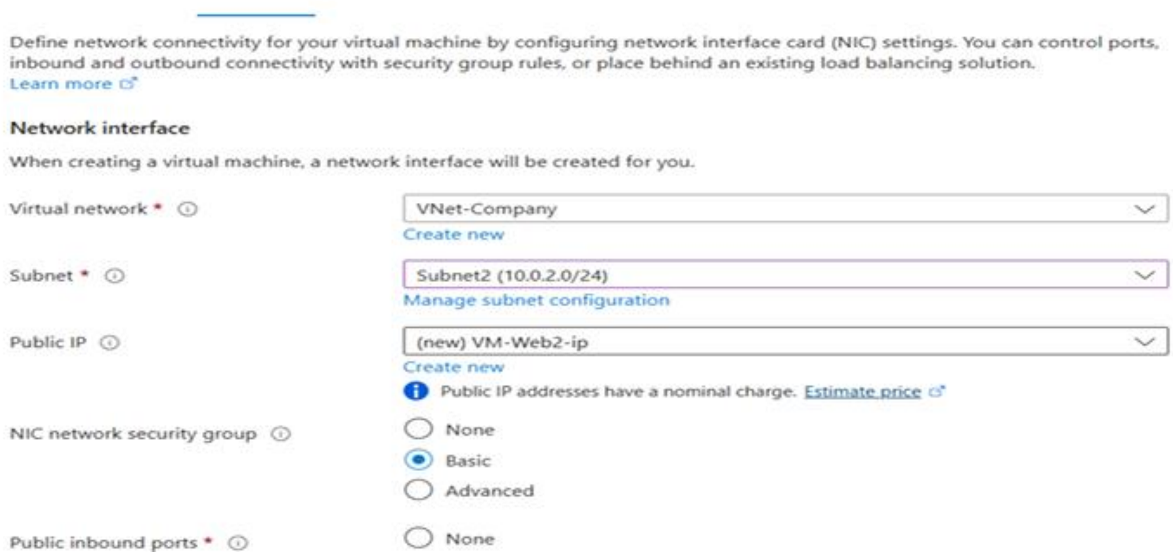
Deploying two web servers in the same Web Subnet in Azure is a best practice for building resilient, scalable, and highly available applications.

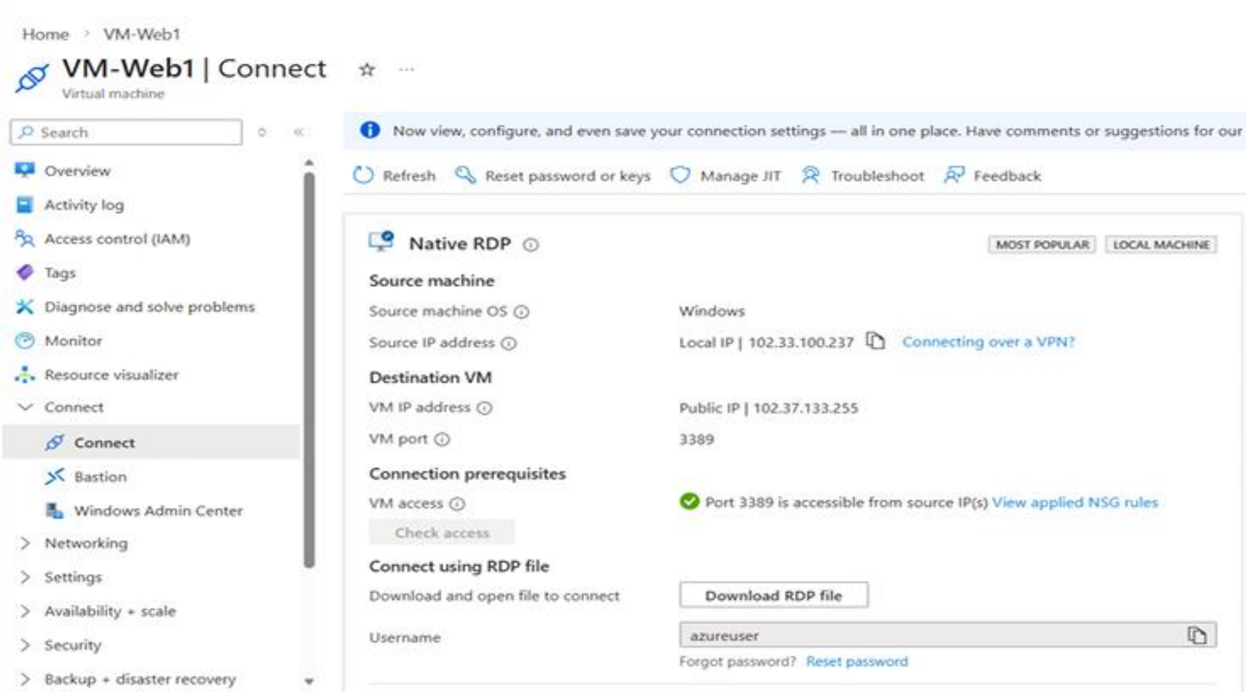


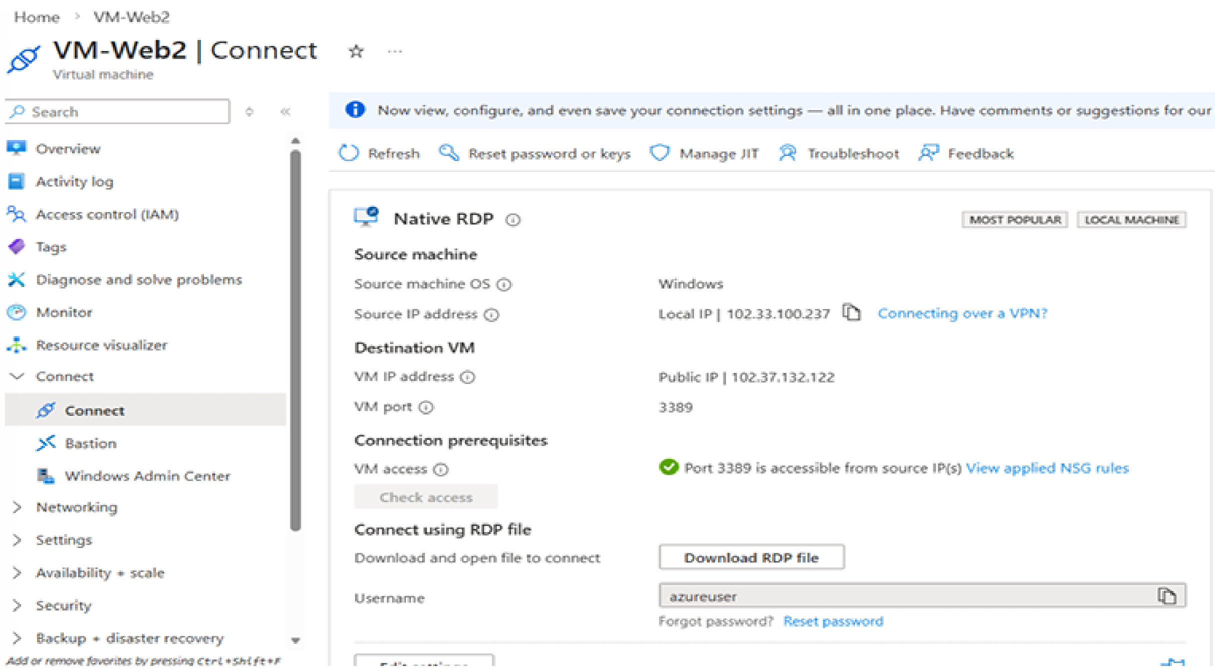


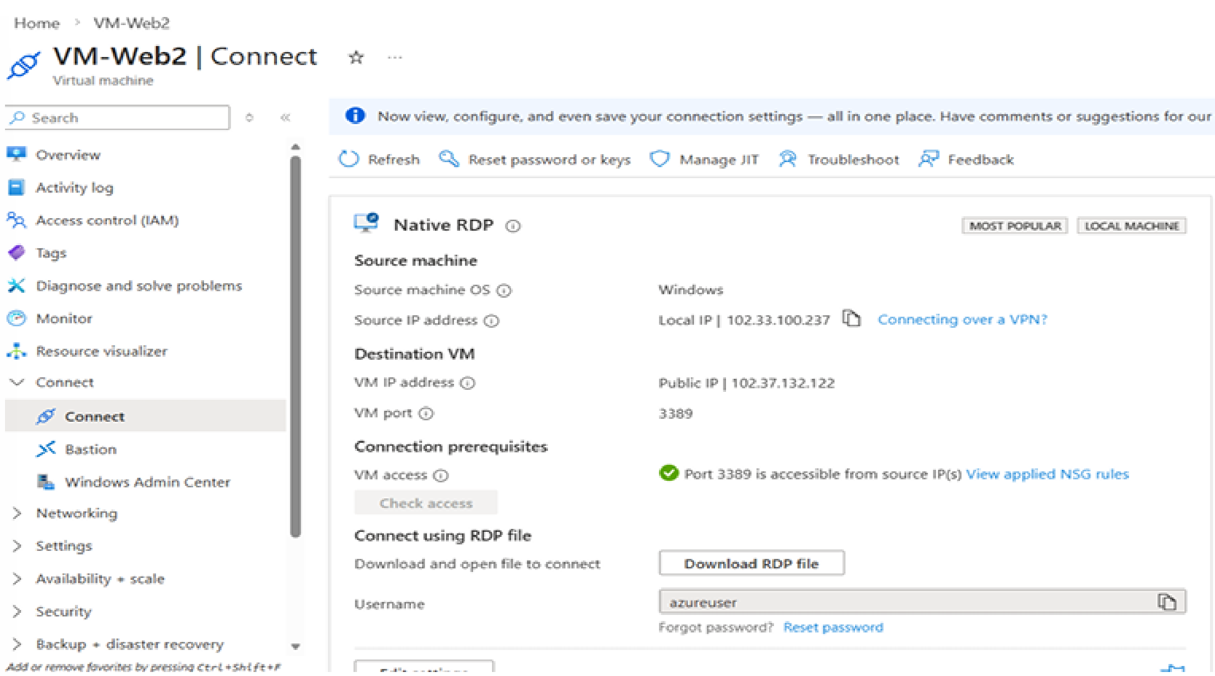


Create two Virtual Machines, named VM-Web1 and VM-Web2, with both using the same image and size. Image = Windows Servers 2025 Datacenter: Azure Edition – x64 Gen2 Size = Standard_B2ats_v2 – 2 vcpus Windows Server 2025 Datacenter: Azure Edition – x64 Gen2 with Standard_B2ats_v2 (2 vCPUs) in Azure is about balancing modern OS features, Azure integration, and cost-effective compute resources. Standard_B2ats_v2 (2 vCPUs) in Azure is suitable for lightweight to medium workloads such as web servers, dev/test environments, or small business apps.






Home > VM-Web2
VM-Web2 | Connect
Virtual machine
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Connect
Bastion
Windows Admin Center
Networking
Settings
Availability + scale
Security
Backup + disaster recovery
Add or remove favorites by pressing Ctrl+Shift+F
Now view, configure, and even save your connection settings — all in one place. Have comments or suggestions for our
Refresh
Reset password or keys
Manage JIT
Troubleshoot
Feedback
Native RDP
MOST POPULAR
LOCAL MACHINE
Source machine
Source machine OS
Windows
Source IP address
Local IP | 102.33.100.237
Connecting over a VPN?
Destination VM
VM IP address
Public IP | 102.37.132.122
VM port
3389
Connection prerequisites
VM access
Port 3389 is accessible from source IP(s) View applied NSG rules
Check access
Connect using RDP file
Download and open file to connect
Download RDP file
Username
azureuser
Forgot password? Reset password


Home > VM-Web2
VM-Web2 | Connect
Virtual machine
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Monitor
Resource visualizer
Connect
Connect
Bastion
Windows Admin Center
Networking
Settings
Availability + scale
Security
Backup + disaster recovery
Add or remove favorites by pressing Ctrl+Shift+F
Now view, configure, and even save your connection settings — all in one place. Have comments or suggestions for our
Refresh
Reset password or keys
Manage JIT
Troubleshoot
Feedback
Native RDP
MOST POPULAR
LOCAL MACHINE
Source machine
Source machine OS
Windows
Source IP address
Local IP | 102.33.100.237
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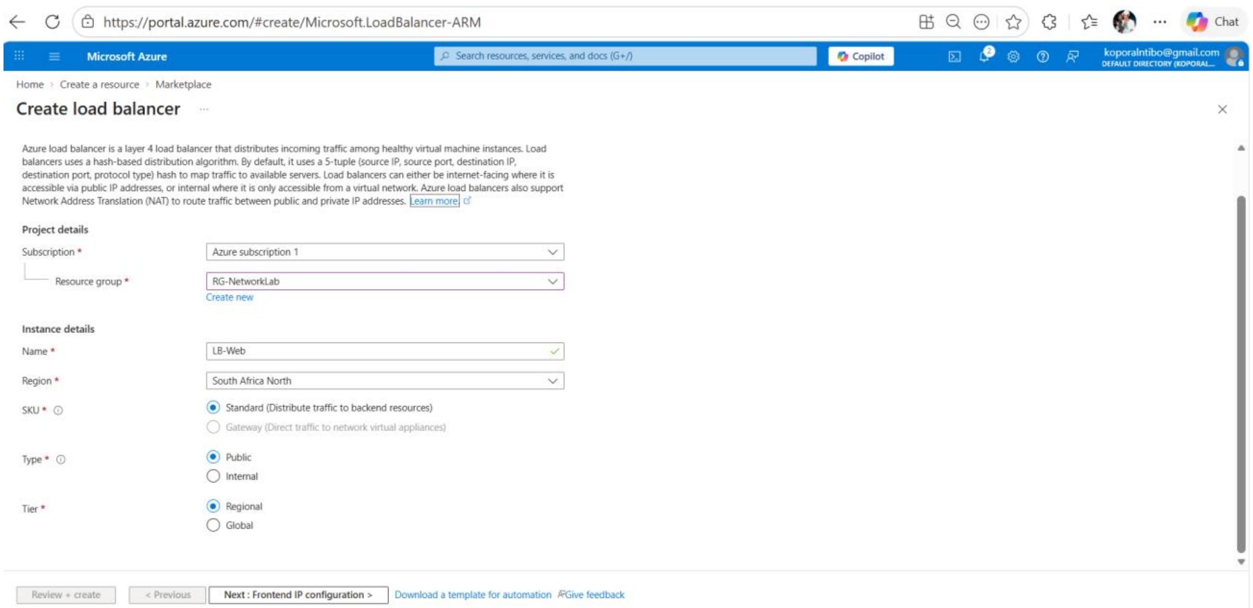
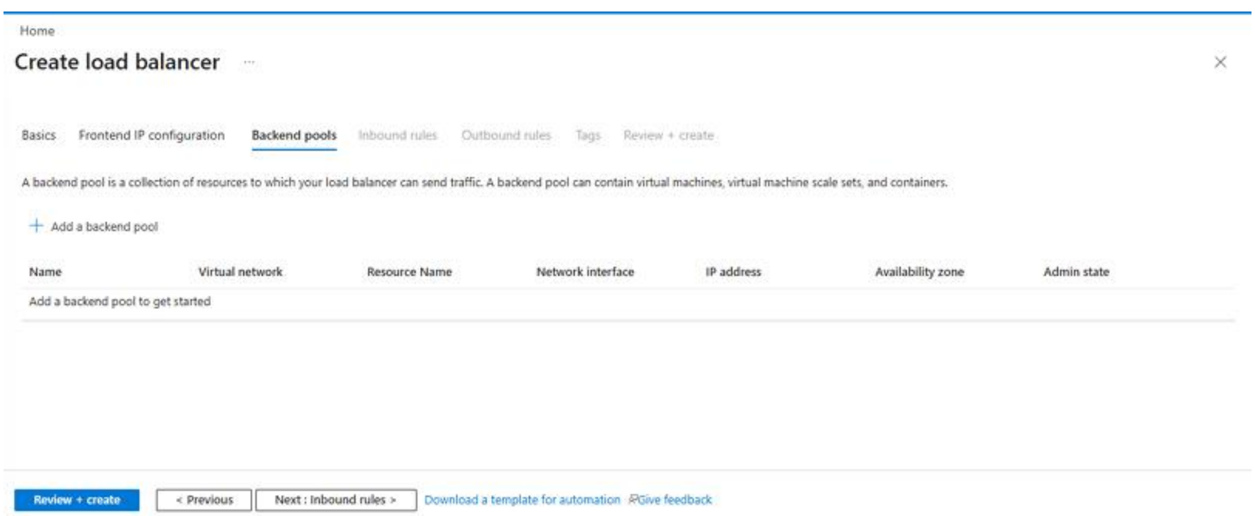
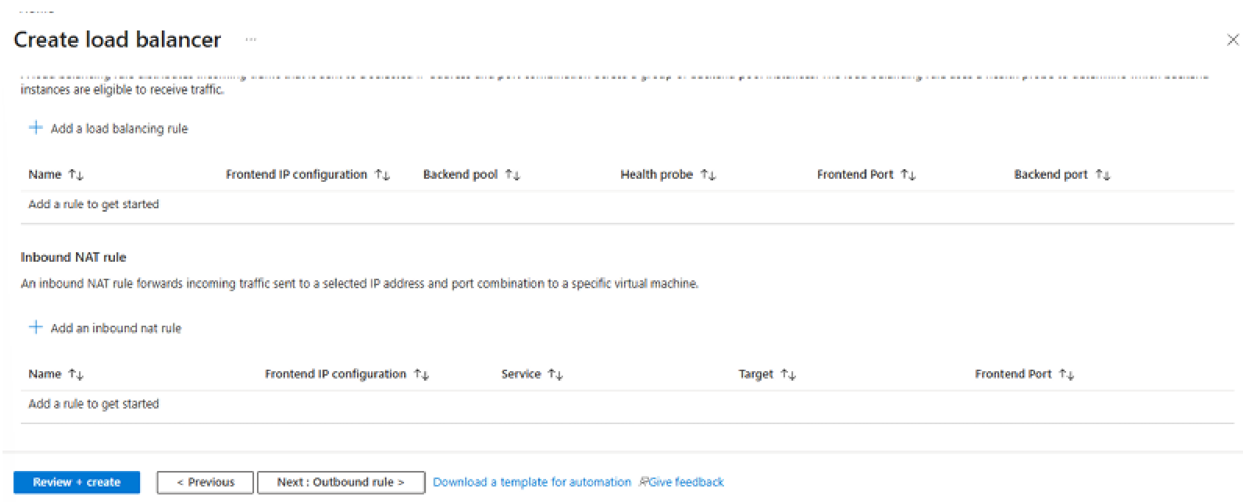
VM access
Port 3389 is accessible from source IP(s) View applied NSG rules
Check access
Connect using RDP file
Download and open file to connect
Download RDP file
Username
azureuser
Forgot password? Reset password

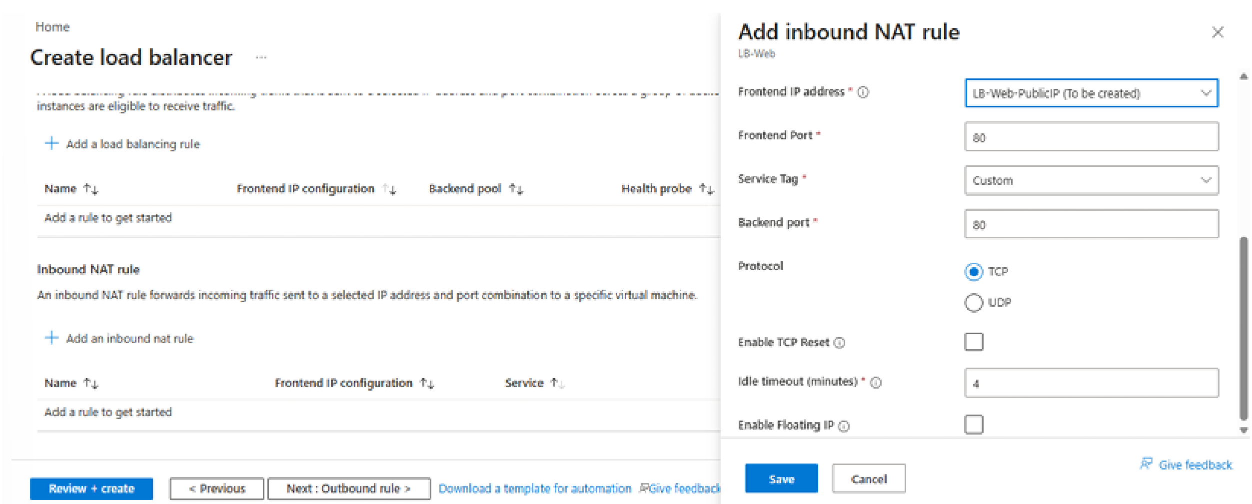
Task 3: Create Load Balancer

Creating a load balancer in Azure is how you distribute traffic across multiple servers or services to ensure high availability, scalability, and performance. The exact steps depend on which type of load balancer you choose (Layer 4 vs Layer 7).

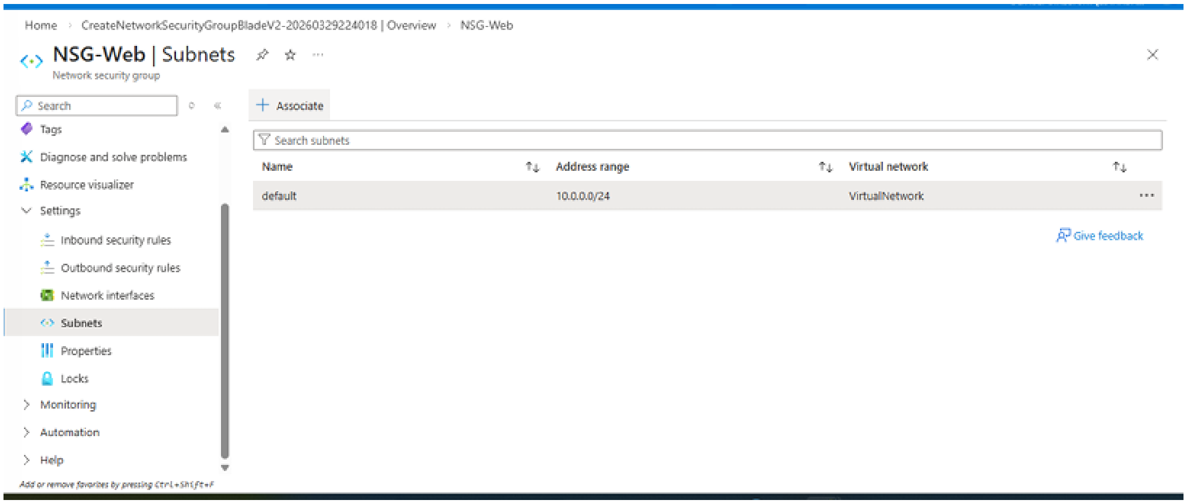





Create load balancer
instances are eligible to receive traffic.
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Frontend Port
Backend port
Add a rule to get started
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Target
Frontend Port
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback


Home
Create load balancer
instances are eligible to receive traffic.
Add a load balancing rule
Name
Frontend IP configuration
Backend pool
Health probe
Add a rule to get started
Inbound NAT rule
An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.
Add an inbound nat rule
Name
Frontend IP configuration
Service
Add a rule to get started
Review + create
< Previous
Next : Outbound rule >
Download a template for automation
Give feedback
Add inbound NAT rule
LB-Web
Frontend IP address *
LB-Web-PublicIP (To be created)
Frontend Port *
80
Service Tag *
Custom
Backend port *
80
Protocol
TCP
UDP
Enable TCP Reset
Idle timeout (minutes) *
4
Enable Floating IP
Save
Cancel
Give feedback

Task 4: Apply Network Security Group (NSG)



A **Network Security Group (NSG)** in Azure is used to control inbound and outbound traffic to resources within a Virtual Network, such as virtual machines, subnets, or network interfaces. To configure the NSG, the basic settings were completed by selecting the same subscription, specifying the resource group, assigning an NSG name, and choosing the appropriate region for the server. Security rules were then added to enhance network protection, including a rule to block Remote Desktop Protocol (RDP) access from the internet by setting the priority to 100, the source to any, the destination port to 3389, and the action to deny. This ensures that unauthorized external access is restricted, allowing only required ports to remain open for secure communication. After completing the configuration and testing, the resource group can be deleted to reduce resource costs and improve overall efficiency in the network.